

Question	Answer	Marks
1(a)	angle (subtended at centre of a circle) when arc (length) = radius	B1
1(b)(i)	$v = r\omega$	C1
	$\omega = 17 / 0.46$ $= 37 \text{ rad s}^{-1}$	A1
1(b)(ii)	$T = 2\pi r / v$ or $T = 2\pi / \omega$	C1
	$= 2\pi \times 0.46 / 17$ or $2\pi / 37$ $= 0.17 \text{ s}$	A1
1(b)(iii)	distance = $2\pi \times 0.038 = 0.24 \text{ m}$	A1
1(b)(iv)	angle = arc length / radius	C1
	$= 0.24 / 0.15$ $= 1.6 \text{ rad}$	A1
1(c)	point X moves through a smaller distance in the same time or (linear) speed of movement of point X / chain decreases or (linear) speed of (circumference of) both cogs decreases	B1
	angular speed (of pedals) decreases	B1

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